What Kind of Water Carves the Mountain?

On the Structure Hidden Inside Different Kinds of Training Data — and Which of It Survives When Precision Is Removed

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A companion to The River and the Canyon, *which built a physical analogy for a transformer — weights as a frozen mountain, activations as water moving over it, training as the slow carving of the stone — and then spent its second half trying to break that analogy on purpose. This paper picks up one question the first one raised but did not answer.*

The first paper treated training as carving and inference as water moving over fixed stone. But it followed only one river: human language. It asked what it means for a model to be shaped by the statistical pressure of text, and was careful to say the resulting terrain was carved by *descriptions* of the world, never the world itself.

This paper asks what changes when the water changes. Code, video, simulation, and physical action do not apply the same pressure prose does; they do not ask a model to preserve the same distinctions, survive the same failures, or discover the same regularities. So the obvious sequel question is: **what kind of water carves the mountain?**

That turns out to be the doorway, not the room. Pursuing it leads somewhere more useful — to a second axis that has nothing to do with where structure came from and everything to do with whether it survives when you take precision away. This paper is about both axes, and about the instrument that exposes the second.

Two Axes, Not One

The temptation, once you notice that different data carves differently, is to build a catalogue: language grooves, code grooves, video grooves, and so on, as though each medium stamps a unique and identifiable signature into the model. That catalogue is real but shallow. The deeper structure has two dimensions, and they are distinct.

Axis 1 — provenance: what carved the structure? Different training media apply different *structural pressures*. They matter not because of their names but because of the kind of structure they repeatedly force a model to learn.

Axis 2 — fragility: how much precision must the structure retain to keep working? Different learned capabilities require different *margins*. Some are broad and forgiving; they survive a coarsened, rounded, less-exact version of themselves. Others are sharp and load-bearing; one bit of drift and the result is not poetically wrong but actually wrong.

The central claim of this paper is that **these two axes are distinct** — fragility is not fixed by provenance. A precise capability learned from prose — careful multi-step deduction, say — can be just as fragile under lost precision as one learned from code. A broad capability learned from code — high-level explanation of what a program does — can be just as robust as casual language fluency. Provenance is an upstream *cause* that shapes the *distribution* of fragile and robust structures a model ends up with. It does not determine the fragility of any particular capability. The compact version, which is also the guard against reading a capability off its data source like a birth chart: **provenance shapes the mix, not the fate.** Keeping the two apart is what makes this a genuine two-dimensional space rather than one idea wearing two labels.

The first paper found the variable underneath the obvious one: it argued that the deep distinction in a transformer is permanence — what training fixes versus what inference leaves transient. This paper does the same move one floor down. The obvious variable is *which data*. The variable underneath it is *which structure, and how fragile* — and the instrument that reveals fragility is one most people treat as a deployment afterthought: quantization.

Axis 1: The Carving Flows

Provenance is best understood not as a set of labels but as a set of pressures, each dominant in a medium without being exclusive to it. (A note on vocabulary, since the first paper used “water” for something subtly different: there, the distinction between training and inference was one of *dosage* — the same water at firehose versus trickle. Here, “what kind of water” is about *composition* — what the training data is made of. The two senses are compatible: a training medium is a particular composition of water, and inference is a low-dose flow of that same mixture. This paper varies the composition; the first varied the dose.)

Language carves description. It is how humans talk about the world, and it is broad, redundant, and forgiving. A vague sentence still parses; a near-synonym still lands. The channels language cuts are wide, which is a strength for coverage and a weakness for precision.

Code carves formal constraint. It forces structure that is exact, nested, long-range, compositional, and state-dependent — and, crucially, externally checkable. Code is language that a formal reality can grade: it runs or it does not, the test passes or it does not, the bracket is closed or the machine refuses. That gradeability is why code is not just another text domain. The clearest evidence that this pressure is real and not decorative is that it *transfers*: pre-training on a mixture of code and text measurably improves general, non-code reasoning — with a reported relative gain of up to 8.2% on natural-language reasoning over text-only baselines — and training on mathematics has shown similar cross-domain benefit into coding and science. The structure code carves is not confined to code.

Video carves temporal continuity. State unfolds through time — objects persist, move, occlude, collide. This is shown rather than asserted, and it carries more world-structure than a sentence describing the same event. There is now direct evidence that this structure is learnable from video alone: self-supervised video models trained without hardwired priors can reliably distinguish physically plausible sequences from implausible ones, acquiring something like object permanence and shape consistency, with above-chance performance from as little as a week of unique video. Intuitive physics, it turns out, need not be built in; it can be carved.

Simulation carves controllable dynamics. Where video shows motion, simulation lets you intervene on it — change a parameter and observe the consequence under rules — and the reason this should carve differently from passive video is not in dispute: intervening and seeing the result sits a rung above mere observation on the causal ladder, the difference between knowing what co-occurs and knowing what causes what. Whether current systems actually acquire that causal structure from interventional training is a live empirical question, with early, single-source signals — one recent result reports that training a model on interventions yields physics knowledge less susceptible to spurious correlation, which transfers to reasoning — but nothing yet that should be called established. Intervention, not just observation, is what *should* carve causal structure; whether it does, at scale, is still being measured.

Action carves consequence. A model whose outputs drive a robot in the world is told *no* by the world when it is wrong, in a way no passive dataset can replicate. This is the top of the ladder and the most speculative rung — but not an empty one: co-training a policy on human video together with robot data has produced capabilities never demonstrated in the robot data alone, such as tidying an unseen home or performing a task with novel structure.

The framing rule throughout: each medium has a *dominant* structural pressure, not a pure one. Language datasets contain mathematics, code, and logic. Code datasets contain a great deal of broken, undocumented, and never-executed code. Video is pixels from a chosen viewpoint, filtered by camera and editing. Simulation is a formalized — and possibly wrong — version of reality. The flows are dominant pressures, not clean categories, and any honest account has to resist the astrology of pretending a capability can be read off its data source like a birth chart.

The Tool That Carves

Before going further, one correction the analogy demands, because without it the whole thesis overclaims. **Water does not carve freely. It carves through a tool, and the tool shapes the cut as much as the water does.**

The same data, fed to a dense transformer, a state-space model, a diffusion model, a mixture-of-experts system, or a robotic policy, will be turned into different geometry. So the claim is not "modality determines structure." The claim is:

> Capability is shaped by the interaction of carving medium, architecture, and training objective — not by data volume alone.

It helps to see architecture as more than the tool held against the stone: it also sets the *grain* of the stone being cut. A dense transformer is closer to uniform granite — every activation passes through every layer. A mixture-of-experts model is more like fractured shale, segmented by router gates, where the path taken depends on which expert a gate selects. The grain matters under coarsening: in granite a trajectory under quantization mostly drifts, but in shale the danger is sharper — a perturbed router can send the computation into the wrong segment entirely, a qualitatively different failure than a small drift within one basin. Same water, different grain, different ways to break.

The objective belongs in that sentence as firmly as the architecture, and it acts on geometry too. Reward only the final answer and you press many precise steps into a single internal transition; reward an explicit chain and you distribute the same work across many visible tokens. This is why chain-of-thought reasoning is more robust under quantization — not because it adds a new skill, but because spreading a computation across externalized intermediate states changes how it is physically implemented, and a distributed implementation has more margin to lose before it fails. A poorly specified objective does the opposite kind of damage: it produces a system that optimizes the letter of its reward while missing its intent — brittle policies, reward hacking, capabilities that look right on the metric and fail off it. Action gives consequence, but the objective decides what "consequence" means. This is the same discipline the first paper applied to the analogy itself: the picture is allowed to be vivid only as long as the mechanism underneath it is kept in view.

Axis 2: Fragility, and the Instrument That Reveals It

Here is the turn that makes this more than a catalogue of data types.

Once a mountain has been carved, you can do something to it that training never does: you can coarsen the stone. You can reduce the precision of the weights — round them, compress them, store them in fewer bits — and then ask which structures still work. This is quantization, and the ordinary way to think about it is as a compression trick, a way to fit a large model onto smaller hardware. That framing misses what it actually is.

Quantization is the inverse problem to training. Training asks what kind of structure a given water can carve. Quantization asks which of those carvings were load-bearing — which parts of the learned geometry require fine resolution to keep functioning, and which are decorative texture that survives being blurred. It is, in other words, the measurement instrument for Axis 2.

And recent evidence shows it is a remarkably precise instrument. Within mathematical and reasoning tasks — the domain where this has been studied most carefully — low-bit post-training quantization does not degrade capability uniformly. It degrades it *selectively*, and the selection is ordered exactly by precision-demand. Harder problems break first: in one systematic study across the Qwen and LLaMA families, aggressive quantization dropped reasoning accuracy by as much as 69.81% in the hardest settings, while leaving easier arithmetic comparatively intact. An earlier paper in the same research line had found up to 32.39% degradation concentrated specifically in numerical computation and planning — the sharp, execution-sensitive operations — while broad conceptual framing survived; the later work extends it with step-level attribution across more benchmarks. So this is one increasingly precise finding rather than independent replication — and what earns trust is not a paper count but the step-localization itself: degradation concentrates in execution, and it concentrates early.

The sharpest finding is finer-grained still. Quantization does not merely break "hard tasks" as wholes. Within a single reasoning chain, it disproportionately elevates *method and execution* errors over *high-level conceptual* errors, and the failure tends to enter early — the first precision-sensitive step flips, and the error cascades to the answer. In the language of this paper, quantization does not wash away an entire trail. It erases the one narrow crossing where the rest of the path depended on a single precise distinction, and the traveler, having lost the footing there, never reaches the far bank. The coarsened model can still describe the route. It can no longer execute the step.

This is the empirical anchor of the whole framework. The fragility axis is not a metaphor hoping for evidence; within reasoning and mathematics it is a measured, step-localized phenomenon. One scoping note matters here: this evidence is about *post-training* quantization of frozen weights — rounding a finished model. It is not a claim about quantization-aware training, mixed-precision schemes, or every choice of calibration data, where a model can be shaped to tolerate coarsening it would not otherwise survive. The instrument this paper leans on is post-hoc coarsening specifically, and a counterexample from quantization-aware training would not refute it; it would be a different question.

This also names a practical stake that is easy to miss. A quantized model can keep its fluent surface behavior while losing the narrow crossings that make its answers actually correct — it still reads as articulate, still frames the problem well, and quietly fails the one exact step the conclusion depended on. Fluency is a wide basin and survives; exactness is a narrow ridge and does not. The risk in deployment is therefore *silent*: the failure does not announce itself as breakdown, because the part of the model that sounds competent is not the part that broke.

The Mechanism Bridge

The first paper insisted, repeatedly, on going down to what the mechanism actually touches — refusing to let a capability-level word stand in for a weight-level fact. The same discipline is required here, because it is tempting and wrong to say "quantization damages reasoning." Quantization does not act on reasoning. It acts on weights and activations. "Reasoning" is a human label for a behavior; the behavior is implemented as some physical property of the numbers, and that property is what coarsening damages. So the bridge has to be built.

The picture the framework proposes is that a model is not one uniform precision substrate. It is a heterogeneous terrain of wide semantic basins and narrow load-bearing ridges. Broad semantic association would live in wide, redundant basins, where many nearby weight settings produce much the same output; rounding moves the state a little but it stays in the basin, so the capability survives coarsening. Exact symbolic manipulation, multi-step state tracking, and numerical execution would live on narrow ridges or near sharp decision boundaries, where the distance between the correct state and its nearest competitor is smaller than the quantization step — so a small rounding pushes the trajectory off the ridge and the answer flips from correct to actually wrong. On this reading, fragility is not a property of "tasks" but of *local geometry*: how wide a margin a capability's implementation leaves before a perturbation changes the output. This should be read as a proposed geometric interpretation of the evidence, not as a measured fact; confirming it directly would mean measuring the margins themselves — the local curvature of the loss landscape, or a capability's activation sensitivity around its decision boundary.

Underneath that geometric reading, there are at least two distinct mechanisms by which a capability can be fragile under quantization, and they can come apart:

The first is **fine-grained spacing**. The capability depends on keeping nearby representational states distinct, and rounding the weights blurs distinctions that needed to stay separate. The model needs *1 + 1 = 2*, not *2.0001*, and the margin between the right state and a near-neighbor is smaller than the rounding error.

The second is **salient or outlier weights**. Quantization error does not spread evenly; it concentrates, because a small minority of weights carry disproportionate signal. The documented finding behind activation-aware quantization is precisely this: protecting roughly one percent of salient weights — identified by measuring activation magnitude, because weight significance is non-uniform — sharply reduces the damage. The complementary finding behind activation-smoothing methods is that activation outliers stretch the quantization range, and migrating that difficulty from activations into weights enables eight-bit quantization of both without accuracy loss. On this mechanism, what degrades is whatever happens to route through the salient channels that low-bit methods sacrifice — which need not coincide with what is "precision-demanding" in the spacing sense at all. It is worth being clear that these outlier channels appear to be substantially an artifact of *architecture* — of how normalization layers interact with certain high-frequency tokens — rather than a mark of the conceptual difficulty of whatever is being computed. They are closer to the model's numerical plumbing than to the geometry of an idea: a high-pressure pipe carrying a disproportionate share of the flow because of how the system was built, not because the water there is doing harder work.

These are different stories, and the honest claim conflates neither. A capability can need fine spacing without depending on outlier channels, in which case ordinary quantization might spare it. A capability can be robust in principle yet fragile in practice because it happens to live in the outlier weights. So the precise statement is not "precision-demanding capabilities degrade." It is: **capabilities degrade to the extent they depend on distinctions quantization cannot preserve — which happens both when the required distinctions are fine-grained, and when they are carried in quantization-fragile weights.** That gives a three-level picture: modality is the upstream cause; precision-demand is the capability-level variable; representational implementation — fine spacing or salient channels — is the weight-level mechanism the instrument actually acts on. The framework earns its keep only when it keeps all three distinct. And because the two failure modes come apart, accuracy curves alone cannot tell them apart: separating them requires localization — which layers and channels lose signal, and whether protecting a salient-weight mask rescues the capability without changing bit depth.

There is a corroborating detail worth noting, because it speaks to the implementation level directly: chain-of-thought prompting makes models *more* robust to quantization. A capability spread across an explicit reasoning chain, with its distinctions externalized into many tokens, is harder to break than the same capability compressed into a single internal step. How a capability is implemented — concentrated or distributed — changes its fragility, independent of what the capability is. (There is a competing reading worth acknowledging: more tokens may simply give the model more chances to self-correct, rather than reflecting a change in where the computation physically lives. The two are not exclusive, and telling them apart is itself a localization question.)

One caveat belongs here, because it is the same wall the first paper named. This whole picture assumes fragility *localizes* — that there is a narrow crossing to find. If a model's features are heavily superposed, smeared across many overlapping directions rather than sitting in tidy separable channels, then the "narrow ridge" will not present as a single clean crossing but as a distributed web of shallow ripples, and degradation under coarsening will look messy rather than sharply thresholded. The cleanliness of the geometric picture is itself gated by how superposed the model is — which is one more reason the instrument needs localization, not just accuracy curves, to say anything precise.

What the Framework Predicts

A framework earns trust by forbidding things. The two-axis picture is not merely a way of organizing what is already known; it makes predictions that could fail, and the strongest of them are *within-modality* — which is what separates this from the shallow claim that each data type leaves a fingerprint.

If fragility tracks precision-demand rather than provenance, then:

Within natural language, exact logical tracking should degrade under quantization before loose summarization does — both are "language," but they demand different sharpness. Within code, high-level explanation of what a program does should survive coarsening longer than exact generation, debugging, or edge-case handling. And within a single code-heavy model, broad programming fluency should persist while exact syntax and long-range dependency tracking fail first.

The sharpest prediction is the one that most directly separates this framework from the shallow "each data type leaves a fingerprint" story: **a model that learned strong formal reasoning from proof-like *text* should show a fragility profile resembling a code-trained model more than a chat-trained one** — because the variable that matters is the precision the structure requires, not the surface form of the data it came from. If that held, provenance-as-fingerprint would be decisively wrong and precision-demand decisively right. The architectural reading of outlier channels sharpens this further into two separable predictions: a capability's *conceptual* fragility — the fine-spacing kind — should track what pressure carved it, while its *systemic* fragility — the outlier-channel kind — should track the architecture and training toolkit instead. They need not move together, and a clean experiment would try to pull them apart.

These are falsifiable. They are supported if capabilities degrade at different bit-depths in a way that localizes to identifiable layers, channels, or activation patterns. They are refuted if quantization sensitivity turns out to be better explained by architecture, model size, calibration data, or layer-norm behavior than by anything about precision-demand or provenance. The within-reasoning evidence already supports the precision-demand half strongly. The provenance half — whether *where a capability came from* predicts *how it breaks* — is the open question this framework exists to pose.

That question is concrete enough to state as a test, and stating it is the framework's main forward-pointing contribution. Take models trained on deliberately different data mixtures — language-heavy, code-heavy, proof-heavy — and quantize each across a range of bit-depths. Then ask not merely whether accuracy drops, but *where* and *whether the pattern differs by provenance* once the obvious confounds are controlled for — architecture, model size, calibration data, task difficulty, and tokenizer, since a code-trained model processes syntax through a different vocabulary distribution and that alone could shift the baseline before any computation begins. Do the fragile capabilities localize to different layers and channels depending on what carved them, and does protecting a salient-weight mask rescue different capabilities in a code-trained model than in a text-trained one? If provenance leaves no signature once those controls are in place, the second axis collapses back toward precision-demand alone, and the framework should say so. The paper does not run this experiment, and as far as we have found, no one yet has — which is precisely why it is offered as the test rather than the result.

What Is Known, What Is Emerging, What Is Speculation

The integrity of a forward-looking paper lives in refusing to state all of its claims at the same level of confidence. Here is the honest ledger.

Established. Three things rest on peer-reviewed, replicated ground. Code-and-text training improves general reasoning, not just coding — the cross-domain transfer is real. Quantization geometry is non-uniform, with a small set of salient weights carrying outsized importance, such that protecting them sharply reduces error. And recursive training on a model's own uncurated synthetic output causes collapse — irreversible loss of the distribution's tails — across model types, a result strong enough to have been demonstrated rather than argued.

Emerging. Several claims have real but still-consolidating evidence. The precision-fragility axis is strongly supported *within mathematics and reasoning*, with the step-level localization described above. Synthetic-data collapse is avoidable when generated data accumulates alongside real data and curation keeps the real fraction above a threshold — so the honest statement is that *indiscriminate* synthetic data collapses a model while *curated synthetic plus real* can work. In the analogy, collapse looks like selective erosion of the low-density tails — the rare, narrow distinctions that uncurated synthetic data no longer replenishes — though it is worth being clear this is the framework's interpretation of the documented effect (tails of the distribution disappearing), not a separately proven claim that collapse targets narrow structure first. Video carves learnable intuitive physics, but does not by itself confer causal understanding: video-language models still struggle to identify physical-law violations, and the field openly asks whether generative video models learn physical principles at all. Simulation, by contrast, shows early promise on exactly the axis video lacks — a recent single-source result reports that training a model on interventions (the general idea being to vary physical parameters and observe the effect, rather than only watching) produces physics knowledge less susceptible to spurious correlation that transfers to reasoning, and the research community has hardened the term "world model" to mean precisely a system that maintains state, exposes action interfaces, and supports counterfactual rollouts, with benchmarks now built to test those capacities falsifiably. Action grounding sits here too: co-training on human video and robot data has produced capabilities never demonstrated in the robot data alone, and the foundation datasets now span dozens of embodiments and hundreds of tasks under a shared action space.

Speculative. Three claims should be stated as open. First and most important: that a capability's *provenance* predicts its *fragility signature* — that code-carved and text-carved structures break differently under quantization in a measurable, predictable way. The precision-demand axis is evidenced; this provenance-by-fragility interaction is not. Second, that self-play or synthetic generation can carve genuinely novel, non-human structure rather than merely amplifying a model's own errors. Third, that physical action and embodiment can close the gap the first paper named — the fact that a language model's terrain was carved by descriptions of the world and never the world itself. These are the frontier, and they should wear that label honestly.

Where the Framework Breaks

A framework, like an analogy, is only trustworthy if it shows where it fails. Several of its joints are weak by construction.

The flows overlap, so provenance is never clean; any claim that reads a capability off its data source is suspect. The tool dominates, so architecture and objective can swamp any effect of the medium. Transfer is the real proof — a modality matters most when training on it improves capabilities *outside* itself, and anything short of that is consistent with the medium mattering far less than it appears. Passive observation is not embodiment: video is not touch, and a model that has watched a million eggs break has not cleaned a counter. Simulation is not reality, and a faithful-looking simulator with the wrong rules teaches wrong structure with great efficiency. Synthetic data can collapse rather than enrich. And, most fundamentally, the framework can frame the search but cannot predict the next architecture — it is a way of asking sharper questions, not a machine for forecasting what will work.

But the most important limit is the one belonging to the instrument itself, and it must be stated plainly because the instrument is the paper's anchor. **Quantization reveals which capabilities are fragile. It does not, by itself, reveal why.** And "why" has two layers it cannot separate unaided. It cannot tell you a capability's provenance — whether the fragile structure came from code or from text. And it cannot, on its own, distinguish the two fragility mechanisms — whether a capability broke because its required distinctions were finely spaced or because it happened to live in the outlier channels that low-bit methods sacrifice. Separating those requires looking at *where* the degradation localizes — which layers, which channels — not merely *that* it occurred. The instrument shows you the fracture. It does not, unaided, tell you the geology that produced it. Naming that blind spot is not a weakness of the framework; it is the condition of using it honestly.

What the Picture Is For

If the two-axis picture holds, two framings follow that are worth stating even though each is a way of reading the evidence rather than a separately proven result. The first: **scale is capacity, not geometry.** More parameters enlarge the total area a model can carve, but they do not by themselves guarantee the narrow ridges that exact capabilities require — fluency can scale while precision lags, until the right pressure is applied. That is the sense in which "more data" is not the whole story, and the structure hidden inside the data matters more than its volume. The second, more speculative: **precision behaves like a learned resource the model allocates.** Broad association gets a redundant, low-precision implementation because it can afford one; exact execution gets a sparse, high-precision one because it cannot afford less. Stated as fact this would overreach — we have not measured any such allocation — but as a reading it captures why the fragility distribution is uneven rather than uniform.

There is a pattern that surfaced repeatedly in assembling this, in places that have nothing obvious to do with one another, and it is worth stating as the framework's most suggestive offering. In low-rank fine-tuning, a frozen base model is grounded by a small, light, removable steering layer. In quantization recovery, a model degraded by compression is restored to near-full capability by fine-tuning on only a few hundred targeted examples. In robotic action-grounding, a model pretrained on a vast amount of cheap human video is grounded into real competence by a *small* amount of true action data. The same shape recurs: a broad, cheap flow builds most of the structure, and a small, targeted dose of a sharper flow makes it load-bearing. If that pattern is real, then the most precious water may not be the most voluminous. The consequence rung may turn out to be less about the *quantity* of action or feedback than about a small amount of exactly the right kind — which is itself a statement about precision, on the second axis, not the first.

And one more complication, offered in the same honest spirit, because it cuts against the tidy image of a ladder. The flows are not strictly separable, and they are not strictly additive. A model can carry deeply carved linguistic and mathematical structure and still possess only shallow physical intuition — the paradox, old in robotics, that the tasks easiest for any animal are hardest for these systems, and the cognition we find most impressive comes cheapest. Yet at sufficient scale and diversity, the flows begin to *fuse*: representations learned from human video and from robot action have been observed to align into a shared, embodiment-agnostic geometry despite enormous surface differences. So the honest picture is neither "separate grooves" nor "one river," but different regions of one terrain that can, under enough pressure, find their way into the same valleys.

Which returns us to the question this paper started from, now visibly the smaller half of a larger one. The first paper asked what a model learns from language. This one asked what changes when the water changes — and found that the more useful question was not *how much* water, nor even *which* water, but which carvings the water leaves are load-bearing, and which wash away the moment the stone is coarsened. Training carves the mountain. Quantization tells us, to the channel, which carvings the mountain actually needs. The provenance of structure and the fragility of structure are two different axes, and the second one — the one we reach by coarsening the stone rather than by sorting the data — is where the sharper questions live.

So the next question is not only what carved the model, but which carved distinctions survive when precision is taken away. The river never truly carved alone; the rock always had its grain, and not every channel it cut was deep enough to keep. Knowing which is which is the whole of the second axis, and most of what is left to learn.

References

The claims in this paper rest on the following work, listed by the section whose claims they support. arXiv identifiers are given so each can be checked directly. Where a claim rests on a single recent preprint, that is noted in the text.

Code-and-text training improves general reasoning. Aryabumi et al., *To Code or Not To Code? Exploring Impact of Code in Pre-training* (arXiv:2408.10914, Cohere, 2024) — reports up to an 8.2% relative increase in natural-language reasoning from adding code to the pre-training mixture, relative to text-only. Ma et al., *At Which Training Stage Does Code Data Help LLMs Reasoning?* (arXiv:2309.16298, 2023) — pre-training on a code-and-text mixture significantly enhances general reasoning almost without negative transfer.

Quantization geometry is non-uniform; a small set of salient weights carries outsized importance. Lin et al., *AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration* (arXiv:2306.00978) — protecting ~1% of salient weights, identified via activation magnitude, sharply reduces error. Xiao et al., *SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models* (arXiv:2211.10438) — migrating difficulty from activation outliers to weights enables 8-bit quantization of both.

Quantization degrades reasoning selectively and step-locally (the empirical anchor). Li et al., *Quantization Meets Reasoning: Exploring and Mitigating Degradation of Low-Bit LLMs in Mathematical Reasoning* (arXiv:2505.11574; Qwen2.5 and LLaMA-3; AWQ/GPTQ/SmoothQuant; GSM8K/MATH/AIME) — degradation up to 69.81% in the hardest settings, with method/execution errors elevated over conceptual errors, failures cascading from an early vulnerable step, and recovery via restoring local margins at the earliest failure frontier. An earlier paper in the same line, Li et al., *Quantization Meets Reasoning: Exploring LLM Low-Bit Quantization Degradation for Mathematical Reasoning* (arXiv:2501.03035) — up to 32.39% degradation (average 11.31%) on Llama-3, concentrated in numerical computation and planning, with recovery from ~545 targeted examples; the 2505 paper extends it with step-level attribution across more benchmarks. *Quantization Hurts Reasoning? An Empirical Study on Quantized Reasoning Models* (arXiv:2504.04823) — degradation worsens with task difficulty and smaller models; W8A8 near-lossless; chain-of-thought increases robustness.

Code quantization is mixed. *Evaluating Quantized Large Language Models for Code Generation* (arXiv:2410.14766) — 4-bit often a good trade-off; very-low-bit degrades.

Quantization preserves fluency more than exactness (supporting the silent-degradation point). *Can Large Language Models Still Explain Themselves?* (arXiv:2601.00282) — finds only moderate declines in self-explanation quality (up to ~4.4%), consistent with surface behavior surviving while exactness erodes.

Video carves learnable intuitive physics but not, by itself, causal understanding. Garrido et al. (Meta), *Intuitive physics understanding emerges from self-supervised pretraining on natural videos* (arXiv:2502.11831) — also the source for the framing of Moravec's paradox here. *TRAVL: A Recipe for Making Video-Language Models Better Judges of Physics Implausibility* (arXiv:2510.07550) — VLMs struggle to identify physics violations. The open question is stated in *Do generative video models learn physical principles from watching videos?* (arXiv:2501.09038).

Simulation / world models: intervention versus observation, and a single-source signal of transfer. The intervention-beats-observation principle is settled causal theory (Pearl's causal hierarchy: observation tells you what co-occurs, intervention what causes). Whether current systems acquire that structure from interventional training is a live question, with an early single-source signal: *Inducing Causal World Models in LLMs for Zero-Shot Physical Reasoning* (CWMI, arXiv:2507.19855). On the generation-versus-understanding distinction and the hardening of the term "world model": *Understanding World or Predicting Future? A Comprehensive Survey of World Models* (arXiv:2411.14499); Warrier et al., *Benchmarking World-Model Learning* (arXiv:2510.19788), which introduces the WorldTest protocol and the AutumnBench suite (43 interactive grid-world environments, 129 tasks across masked-frame prediction, planning, and causal-dynamics-change detection; humans substantially outperform frontier models). On the architectural constraint that physical causality requires the present to depend only on the past: *LingBot-VA* (arXiv:2601.21998).

Action grounding yields capabilities not present in robot data alone; flows fuse at scale. *Emergence of Human to Robot Transfer in Vision-Language-Action Models* (arXiv:2512.22414) — co-training on human video and robot data produces capabilities never shown in robot data, and embodiment-agnostic representations emerge with pretraining diversity. Foundation work: *Open X-Embodiment* (Collaboration et al., 2023; 22 embodiments, 500+ tasks, shared action space), RT-1 (arXiv:2212.06817), and OpenVLA (Kim et al., arXiv:2406.09246; 970k trajectories, outperforming the 55B RT-2-X). On a small amount of action data grounding broad video pretraining: arXiv:2510.21571.

Model collapse from recursive synthetic training, and its avoidability. Shumailov et al., *AI models collapse when trained on recursively generated data* (*Nature*, 2024; doi:10.1038/s41586-024-07566-y) — uncurated recursive synthetic training causes irreversible loss of the distribution's tails, across model types. Gerstgrasser et al., *Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data* (arXiv:2404.01413, 2024) — collapse is avoided when synthetic data accumulates alongside real data rather than replacing it.

This is the first draft of a companion to The River and the Canyon. *It inherits that paper's analogy and its discipline: the picture is a way of asking sharper questions, not a machine for predicting architectures. Where it names established results it has tried to be accurate; where it reaches toward the frontier it has tried to say so plainly. The margin-geometry reading of fragility is offered as an interpretation consistent with current evidence, not as a measured fact. Written by E. A. Flores, Apiana AI, Inc.*